

Mikhail Razakov

Compiler Engineer — LLVM / Backend / Code Generation

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Compiler engineer with professional LLVM 22 experience at MCST. Public projects demonstrate conservative LLVM transformations and a separate x86-64 backend with SSA/CFG validation, liveness/interference, register allocation, spills, phi lowering, and native-vs-interpreter differential execution.

EXPERIENCE

- 2026 — present** **ISP RAS — Static Analysis Engineer**
- Contribute to SharpChecker, ISP RAS's industrial static-analysis platform for C#.NET.
- July — August 2026** **MCST — Compiler Engineering Intern**
- Implemented an LLVM 22 LICM pass with loop analysis, side-effect/speculative-safety checks, and hoisting only proven invariants to preheaders.
 - Developed a conservative interprocedural global-IV prototype with APInt affine evolution, transitive call effects, a separate legality phase, and LLVM IR transformation; unsupported cases are rejected unchanged.

PROJECTS

LLVM Interprocedural Global IV Optimization

An LLVM 22 pass localizes supported induction-like globals after interprocedural effects/affine analysis and a separate legality decision; APInt models fixed-width wraparound.

29 positive/negative regression cases: global-iv, verify, structural expectations, idempotence, and observable before/after execution comparison.

x86-64 Codegen & Register Allocation Lab

A compiler backend with SSA/CFG validation, liveness/interference analysis, linear-scan and seeded simulated-annealing allocators, spills, phi lowering, and a SysV x86-64 emitter.

An independent assignment verifier plus native-vs-interpreter differential execution; tests cover branches, loops, spills, and parallel phi moves.

UniversalToolchain

Modular compiler/runtime infrastructure: typed artifact contracts, dependencies/conflicts, provider selection, pass ordering, and artifact routes compile into an immutable LanguagePlan before execution.

Current exact test manifest: 1,324/1,324 passing; interpreter/CIL parity, exact package binding, and fail-closed runtime checks protect the selected execution plan.

SKILLS

Compiler / IR

C++23, LLVM 22/IR, APInt, CFG/SSA, dominators, loops, LICM, interprocedural analysis, legality

Backend / Codegen

SysV x86-64, liveness/interference, register allocation, linear scan, simulated annealing, spills, phi lowering

Analysis / Correctness

LLVM Verifier, conservative rejection, structural regression checks, idempotence, differential execution

Systems / Tooling

Rust, CMake, Linux, Python harnesses, low-level debugging

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

LangDev'26 — Build the Language, Then Make the Abstractions Disappear

Accepted talk on UniversalToolchain's architecture; LangDev'26 takes place October 8–9, 2026 in Málaga.

UniversalToolchain — Baltic Science and Engineering Competition

First Degree Diploma and Grand Prize "Perfection as Hope" for UniversalToolchain.

MEPhI Junior

Overall winner; First Degree Diploma, 96/100 for UniversalToolchain.

Moscow / remote